Example 4.27: Solve $\tan^{-1}2x+\tan^{-1}3x=\frac{\pi}{4}$, if $6x^2<1$.

Solution

Now, $\tan^{-1}2x+\tan^{-1}3x = \tan^{-1}\left(\frac{2x+3x}{1-6x^2}\right)$, since $6x^2<1$.

So, $\tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{\pi}{4}$, which implies $\frac{5x}{1-6x^2}=\tan\frac{\pi}{4}=1$.

Thus, $1-6x^2 = 5x$, which gives $6x^2+5x-1=0$

Hence, $x = \frac{1}{6},-1$. But $x=-1$ does not satisfy $6x^2<1$.

Observe that $x=-1$ makes the left side of the equation negative whereas the right side is a positive number. Thus, $x=-1$ is not a solution. Hence, $x=\frac{1}{6}$ is the only solution of the equation.

Example 4.28

Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$.

Solution

Now, $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right) = \tan^{-1}\left[\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\frac{x-1}{x-2}\left(\frac{x+1}{x+2}\right)}\right]=\frac{\pi}{4}$.

Thus, $\frac{\frac{x-1}{x-2}+\frac{x+1}{x+2}}{1-\frac{x-1}{x-2}\left(\frac{x+1}{x+2}\right)} = 1$, which on simplification gives $2x^2-4=-3$

Thus, $x^2 = \frac{1}{2}$ gives $x=\pm\frac{1}{\sqrt{2}}$.

Example 4.29

Solve $\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \sin\left\{\cot^{-1}\left(\frac{3}{4}\right)\right\}$.

Solution

We know that $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)=\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$.

Thus, $\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \frac{1}{\sqrt{1+x^2}}$. ...(1)

Let $\cot^{-1}\left(\frac{3}{4}\right)=\theta$. Then $\cot\theta=\frac{3}{4}$ and so θ is acute.

From the diagram, we get

Hence, $\sin\left\{\cot^{-1}\left(\frac{3}{4}\right)\right\}=\sin\theta=\frac{4}{5}$ ... (2)

Using (1) and (2) in the given equation, we get $\frac{1}{\sqrt{1+x^2}} = \frac{4}{5}$, which gives $\sqrt{1+x^2}=\frac{5}{4}$

Thus, $x=\pm\frac{3}{4}$.

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